

GL-DirL-en-001

Product E2E Traceability Directive

Process Owner: OLGA MONDRIIEVSKA

Approver 1: GURVINDER PAL SINGH

Approver QMS: ERIC JACQUOT

Revision: 1.0

Effective Date: 03/26/2025

Revision Date: 03/05/2025

Life Is On



Purpose

The purpose of this Directive is to define the requirements to implement traceability for any new or legacy products manufactured In or For Schneider Electric and part of the “critical offer”. To guide entities to implement traceability on their perimeter of accountability.

Scope

Requirements in this Directive apply to all manufacturing plants, manufacture Finished Goods and Sub-Assemblies, and distribution centers supply to the final Outside Group Customer or other SE internal entities part of global supply chain process.

Target Audience

- CS&Q
- Industrialization
- Procurement
- Logistics
- Products Owners/Offer managers

Table of Contents

Topic	See Page
References	2
Process Inputs	2
Process Overview	2
OG/IG components/Sub-Assemblies serialized with GS1 standard	3
Finished Good serialization with MR064 standard	4
Traceability data record and storage within manufacturing process	4
Manufacturing plant MES with traceability database ALM digital connection	4
Logistics traceability process	5
Process Outputs	5
Monitoring & Control	5
Roles & Responsibilities	6
Competency Requirements	6
Quality Records	6
Terms & Acronyms	7
Document Control	7
Appendix 1: E2E Traceability process map	8
Appendix 2: MES ALM connection Data Model	9
Appendix 3: E2E Traceability ERP Process flow for Plant and DC	10

Product E2E Traceability Directive, Continued

References

- FTJ01003 Global Traceability Directive and Guideline (link to [System Interoperability Library](#))
 - FTJ01002 Guide Data Matrix Label requirements (link to [System Interoperability Library](#))
 - MR064 Product Identification from QR Code and S/N Coding (link to [System Interoperability Library](#))
 - TI114 QR Code Definition (link to [System Interoperability Library](#))
 - [E2E Traceability page](#)
-

Process Inputs

- Critical offers as per LoB definition
 - Critical components/Sub Assemblies/Finished Goods
 - Standard MES/upgraded legacy MES
 - Manufacturing/Logistics traceability data
 - E2E Traceability requirements with respect to FTJ01003/FTJ01002/MR064/TI114
-

Process Overview

The Offer E2E Traceability (see process map in [Appendix A](#)) includes:

- [OG/IG components/Sub-Assemblies serialized with GS1 standard](#)
 - [Finished Good serialization with MR064 standard](#)
 - [Traceability data record and storage within manufacturing process](#)
 - [Manufacturing plant MES with traceability database ALM \(Asset Lifecycle management\) digital connection](#)
 - [Logistics traceability process](#)
-

Product E2E Traceability Directive, Continued

OG/IG
components/Sub-Assemblies
serialized with
GS1 standard

All products part of “Critical Offers” list defined by LoB shall be traced. To know if the entity has to set up traceability, refer to Products Owners/Offer managers.

General rules shall be applied for Critical components serialization:

- Batch serialization for critical part is minimal requirement
- Part is defined as Critical following Technical Folder categorization (Critical, Major, Minor) and its criteria is managed by Continuous Engineering
- Entity can extend this requirement of traceability if needed with other parts
- Batch serialization is done through label stick on each packaging delivered to entity
- Serialization shall be compliant with FTJ01002
- All the identified critical components shall be defined with “Default trace code” in PDM

It’s **LoB** responsibility to deliver and keep up to date Critical Parts list in Symphony/PDM required to be traced and recorded in MES in manufacturing plant.

It’s **Entity responsibility** to link Critical Part batch number with Finished goods and record this link in MES.

It’s **Supplier** responsibility to deliver Critical Parts with traceability compliant to specification on box and record for these parameters and relevant data for it.

FTJ01002 for component serialization rules are universal rules. Any old serialization guideline shall be aligned/replaced by FTJ01002.
Please refer to FTJ01002 Guide Data Matrix Label requirements

Product E2E Traceability Directive, Continued

Finished Good serialization with MR064 standard

All Finished Good as part of “Critical Offers” defined by LoB shall be serialized with MR064.

Finished goods serialization rules:

- Finished goods bare product shall be serialized based on the marking rule “MR064”
- Product individual packaging shall be serialized with 2D DataMatrix with product serial number content only

Refer to FTJ01003 Global Traceability Directive and Guideline and MR064 Product Identification from QR Code and S/N Coding.

Traceability data record and storage within manufacturing process

Each SE Entity manages and records traceability for the process in its perimeter, starting from critical components to finished good product. Traceability of Critical Offer is recorded in standard MES Quality modules – PMS, iTAC, LDS E-Quality.

In exception where the standard Quality module is not used, but a local legacy MES is used due to unavoidable circumstances (ex. Legacy MES linked with several other operations of factory, other justifiable reasons etc), it’s the responsibility of that entity to ensure that the legacy MES is certified, compatible and adheres to manage and ensure the traceability needs.

All requested traceability data shall be managed and stored with rules as per FTJ01003 Global Traceability Directive and Guideline.

Manufacturing plant MES with traceability database ALM digital connection

All requested traceability data shall flow to ALM (Asset Lifecycle Manager).

There is a technical clarification on the fields and values which are mandatory and optional required to be sent to ALM as manufacturing traceability data summarized and simplified as in [Appendix 2](#) MES ALM connection Data Model.

Refer to FTJ01003 Global Traceability Directive and Guideline section 2.2. for **ALM Input Data file**.

Product E2E Traceability Directive, Continued

Logistics traceability process

To ensure E2E traceability of the Finished goods throughout entire SE GSC to Outside group customer, factory and DC traceability solutions shall be deployed for respective entities.

Serial number scanning process shall be put in place for all Finished goods products defined under traceability scope in order to link serial number with customer order.

Refer to [Appendix 3](#) E2E Traceability ERP Process flow for Plant and DC.

Process Outputs

- OG/IG Critical components/sub-assemblies serialized with GS1 standard
 - Finished Good serialized with MR064 standard
 - Critical traceability data (refer to FTJ01003 Global Traceability Directive and Guideline section 4.1 for critical traceability data record)
 - All “critical offer” Finished Good product Serial Number linked with IG/OG Customer shipping information
-

Monitoring & Control

E2E Traceability deployment is part of Manufacturing and Logistics SPS audit.

KPIs to monitor include:

- 100% of MES ALM connection for E2E Traceability defined scope
 - OG/IG components/Sub Assemblies serialization with GS1 standard
 - Digital logbook deployment as per product scope defined for each plant
 - %, manufactured Finished Good quantity available in ALM/Finished Good qty produced-dispatched from plant
 - %, serial number scanning activation
 - %, quantity serial numbers in ALM with good data quality/ quantity of serial numbers scanned available in ALM
 - %, quantity serial numbers available in ALM/quantity serial numbers shipped
-

Product E2E Traceability Directive, Continued

Roles & Responsibilities

Role	Responsibilities
Business Unit (BU)	E2E Traceability "Critical" offer definition
	Critical Components & CTP/CTQ definition
	Finished Good SKU list definition
Procurement/GSC plants	GS1 deployment for Critical Components OG
CS&Q/ GSC plants	GS1 deployment for Critical Components IG
CS&Q/ Open IT/Plant Industrialization	Standard MES deployment/legacy MES upgrade (as per Industrial guidelines)
CS&Q/Plant Industrialization	MES connection to ALM
GSC plant	Aggregation code deployment on secondary packaging labels
Business Unit (BU)/Plant Industrialization (deployment)	Finish good Serial Number compliance as per MR064
	QR code deployment on the bare product
	Serial Number Data Matrix on primary packaging label
Business Unit (BU)/GSC Logistics	Serial Number activation in ERPs
Digital/GSC plant	ERP connection with ALM

Competency Requirements

Refer to [SoR](#) (System of Reference: Job Codes & Competencies), [LoP](#) (Levels of Proficiency) & [CoMET](#) for detailed competencies.

Quality Records

Record	Record Location	Retention Period
Critical traceability data	Plant MES Quality module	From 10 to 25 years, following manufacturing date of Finished Good
Critical traceability data record	Traceability data base ALM (via MES link)	No retention period limit
Shipped Finished Good Serial Number data record with IG/OG Customer shipping information	Plant& DC ERP/ ALM (Asset Lifecycle management)	NA

Product E2E Traceability Directive, Continued

Terms & Acronyms

The table below includes definitions for terms and acronyms used in this document. Reference [ISEE](#) for a full list of terms and acronyms used in the SE IMS.

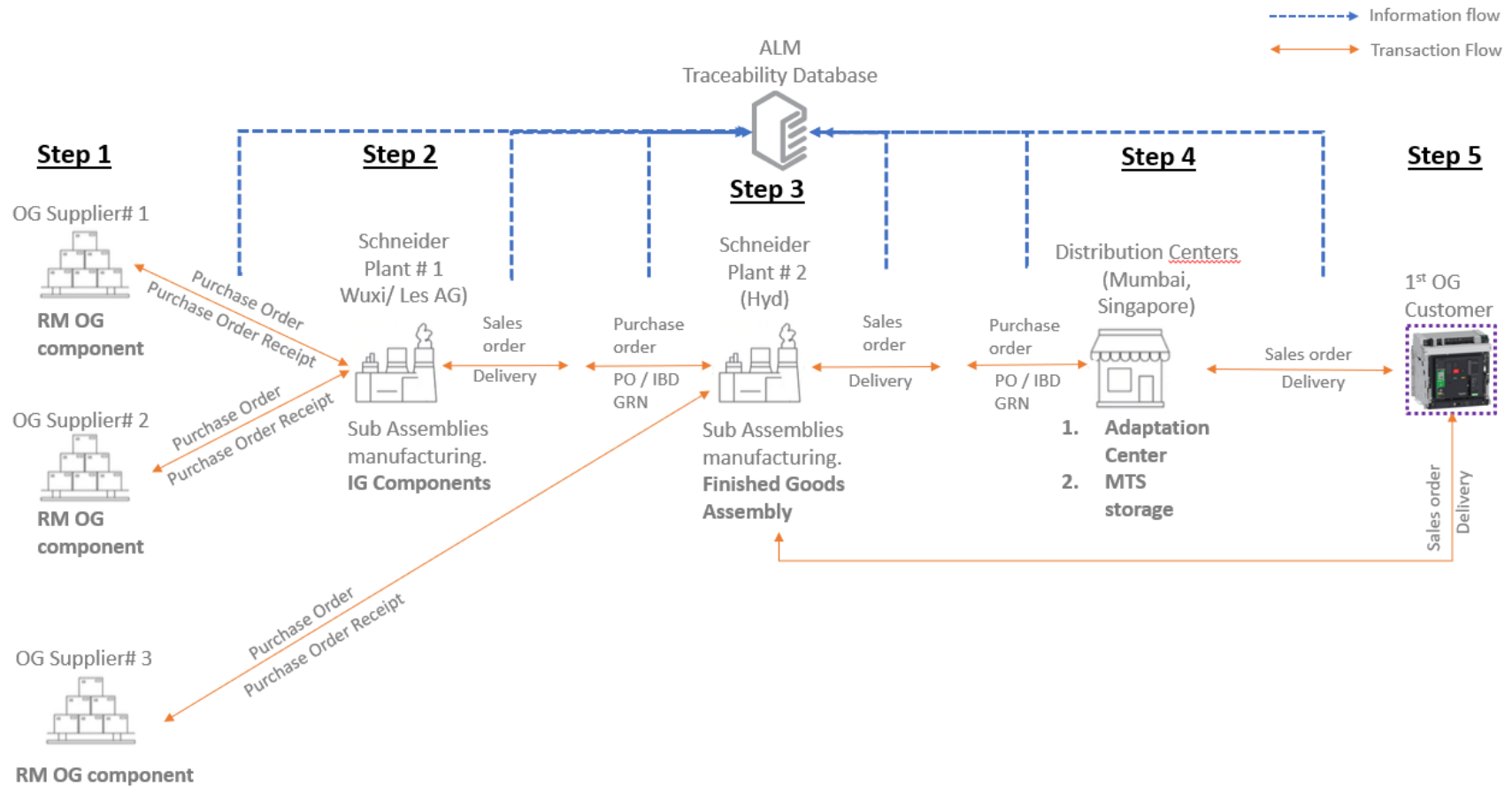
Term	Definition
ALM	Asset Lifecycle management
LoB	Line of Business
BU	Business Unit
CS&Q	Customer Satisfaction and Quality
MES	Manufacturing Execution System
SKU	Stock Keeping Unit
OG	Outside Group
IG	Inside Group
FG	Finished Good
DC	Distribution Center
DM	Data Matrix
CTP	Critical to Process
CTQ	Critical to Quality
PDM	Product Data Management

Document Control

Function	Name
Process Owner	Olga Mondrievska
Verifier/ Functional Leader	Gurvinder Pal Singh
Approver/ CS&Q QMS VP	Eric Jacquot

Revision	Date	Author	Summary of Change
1.0	March 5th, 2025	Olga Mondrievska	Initial release.

Appendix 1: E2E Traceability process map



Appendix 2: MES ALM connection Data Model

Product Table				Component Table			
Product Serial Number	String	M		Product Serial Number	String	M	
Product Commercial Reference	String	M		Product Commercial Reference	String	M	
Product Version <i>Evolution of the product</i>	String	O		Component Reference Number	String	M	
Software Version <i>Firmware version embeded in the product</i>	String	O		Component Traceability Number (serial number or batch number)	String	M	
Product Manufacturing Date	Date	O		Type of traceability ID - S (for Serial) or B (for batch)	String	O	
Manufacturing Order Number	String	O		Component Manufacturing Date	Date	O	
Manufacturing Order Start TimeStamp	DateTime	O		Component Standard Supplier ID (DUNS)	String	M	
Manufacturing Order Close TimeStamp	DateTime	O		Component Version	String	O	
Standard Plant ID <i>As per Manufacturing Data standard</i>	String	M		Quantity	Number	O	5
Standard Production Line ID <i>As per Manufacturing Data standard</i>	String	O		Units	String	O	
Certificate key <i>Only needed when the product has a firmware</i>	String	O					
Made In <i>As marked on the product</i>	String	O					

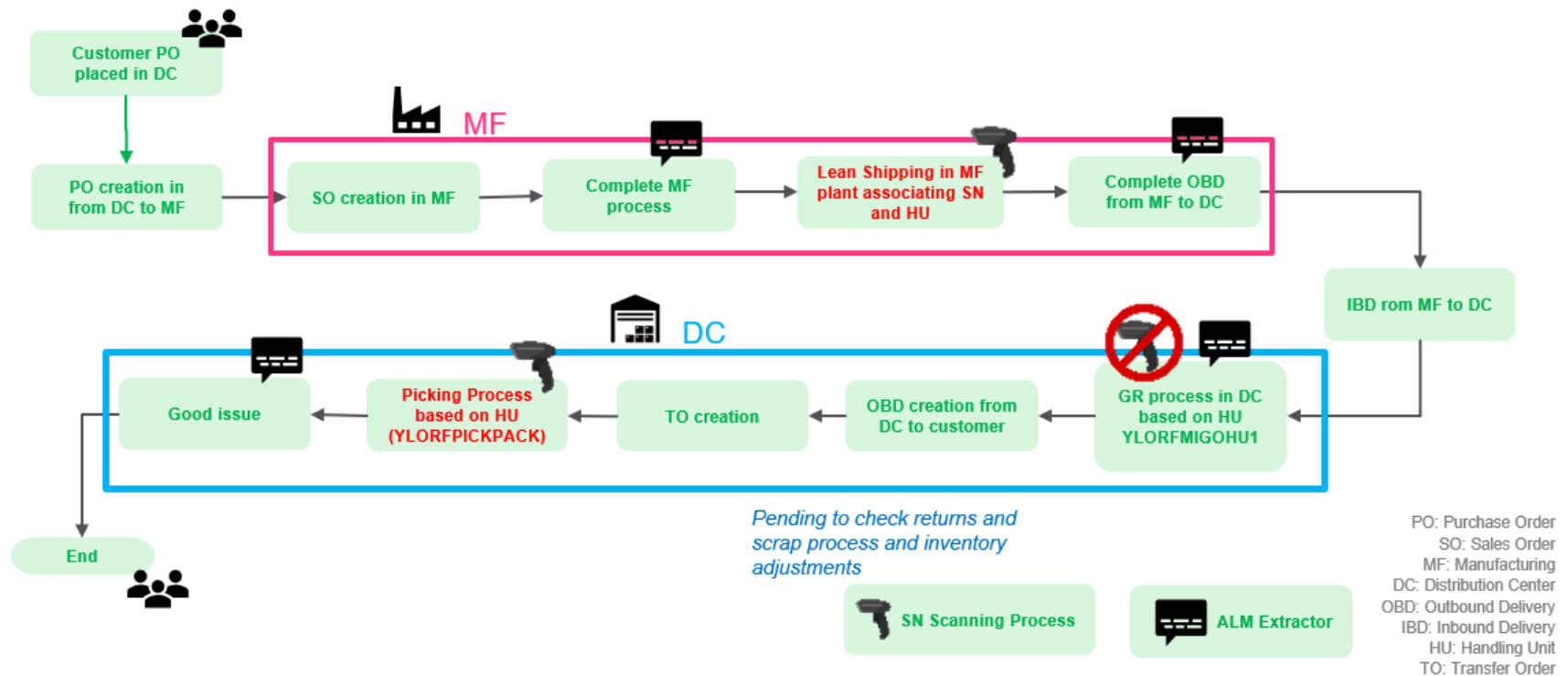
***M** – mandatory data to be sent to ALM

O – optional data to be sent to ALM

Manufacturing Date format example: 2021-05-03T11:39:07.653Z

Product manufacturing date forma could be derogated: 2021-05-03

Appendix 3: E2E Traceability ERP Process flow for Plant and DC



REVISION HISTORY SUMMARY

Approvers	Requested By	Change Description	Reason for Change	Revision	Revision Date	Contributors
ERIC JACQUOT; GURVINDER PAL SINGH	OLGA MONDRIIEVSKA	* This document has been migrated from the Global QMS Documents Platform (SharePoint) without revision changes. Process owner/Document author and revision details are specified in the document itself. * A summary of previous revisions is available only in the migrated version.	Document migration	1.0	03/05/2025	